Source Aware

Retrieval-Augmented Generation with Estimation of Source Reliability
Jeongyeon Hwang et al. (2024/2025)

Paper: https://arxiv.org/abs/2410.22954

What the paper proposes

Reliability-Aware RAG argues that semantic similarity alone ignores heterogeneous source trustworthiness and proposes reliability-aware source selection and aggregation.

What changed in this project

Applied an explicit provenance prior to dense scores: official 1.08, technical 1.03, unknown 1.00, community 0.96. This is isolated from the application pipeline and uses the existing source_type metadata.

Measured results

Metric	Baseline	Method	Change
Hit@1	80.0%	80.0%	+0.0 pp
Hit@5	85.0%	85.0%	+0.0 pp
Recall@5	64.7%	64.7%	+0.0 pp
NDCG@10	84.0%	84.0%	+0.0 pp
MRR@20	82.9%	82.9%	+0.0 pp
Latency	26.9 ms	10.4 ms	—

Inference

This corpus is almost entirely official manufacturer documentation, so the prior did not change aggregate metrics: Hit@5 remained 85.0%, Recall@5 64.7%, NDCG@10 84.0%, and MRR@20 82.9%. It is a useful limitation finding: source-aware ranking needs heterogeneous sources to demonstrate value.